

# Discrete Hexagonal Differential Forms: Metric Boundary Quantization and Thermodynamic Invariants on Discrete Global Grid Systems

Web of Life Research Collective

<https://github.com/pascalranoroarijaona/WebOfLife>

Sprint 051 Technical Report

February 2025

## Abstract

Planetary-scale ecological and geophysical simulations require computational discretizations that rigorously preserve fundamental thermodynamic balance laws while resolving non-equilibrium dissipative fluxes. While Uber’s H3 Discrete Global Grid System (DGGS) offers an equal-area, low-distortion tessellation of the terrestrial sphere, numerical transport schemes often collapse shared polygonal boundaries to idealized point-to-point connections. Such approximations omit shared geodesic edge lengths, directional contact normal vectors, variable boundary cross-sections, and topographic slopes, producing artificial numerical dispersion, anisotropic dissipation errors, and thermodynamic violations. In this paper, we present the theoretical derivation and software contract of `H3CellInterfaceMetrics`, implemented in `src/spatial/h3_types.ts` within the Web of Life simulation engine. We establish discrete finite-volume differential operators over H3 cells that capture subsurface Porous (Darcy-Richards) flow, Saint-Venant overland runoff, atmospheric boundary-layer turbulence, and advective nutrient-carbon dynamics. We formally prove that this metric boundary quantization guarantees exact First Law mass and enthalpy skew-symmetry ( $\Phi_{i \rightarrow j} = -\Phi_{j \rightarrow i}$ ) alongside Second Law positive-definite entropy generation ( $\dot{S}_{\text{gen}} \geq 0$ ).

## 1 Introduction

Simulating the Earth’s biosphere across spatial and temporal scales demands a non-equilibrium thermodynamic perspective [1, 2]. Planetary life is not an isolated mechanical aggregate; it is an open dissipative structure organized through interconnected cascades of solar exergy dissipation, water mass cycles, biogeochemical transport, and thermodynamic entropy generation [3, 4].

A long-standing challenge in digital biosphere modeling is spatial discretization over the spherical Earth [5]. Standard latitude-longitude grids introduce polar coordinate singularities and extreme area distortion, requiring heavy polar filtering that destroys local conservation invariants. Uber’s H3 Discrete Global Grid System (DGGS) circumvents these coordinate singularities by projecting an icosahedral aperture-7 hexagonal tessellation onto the terrestrial ellipsoid.

However, resolving continuous conservation equations on an H3 tessellation requires moving beyond naive graph-network approximations where cells are treated as dimensionless point nodes connected by isotropic edges. In real fluid and porous transport:

1. Shared boundaries between adjacent Voronoi cells possess finite geodesic lengths ( $L_{ij}$ ) that vary across the icosahedron faces and pentagonal disclinations.
2. Directional advection depends on the inner product between fluid velocity fields  $\vec{u}$  and the unit normal vector  $\hat{n}_{ij}$  of the shared facet.

3. Subsurface hydrology, overland runoff, and atmospheric boundary-layer exchange occur through distinct physical vertical layers, requiring distinct stratified cross-sectional contact areas ( $A_{ij}^{\text{sub}}, A_{ij}^{\text{surf}}, A_{ij}^{\text{atm}}$ ).
4. Topographic gradients along the interface normal direct gravity-driven hydraulic potentials.

To establish a thermodynamically sound foundation for spatial transport, Sprint 051 formalizes the `H3CellInterfaceMetrics` boundary metric specification within the Web of Life project (<https://github.com/pascalranoroarijaona/WebOfLife>). This paper details its mathematical formulation, thermodynamic invariance proofs, and software integration.

## 2 Finite-Volume Discretization on DGGS

Let  $\mathcal{M}$  denote the two-dimensional spherical manifold representing the Earth's surface. The DGGS partitions  $\mathcal{M}$  into a set of discrete, non-overlapping polygonal cells  $\mathcal{C} = \{c_i\}_{i=1}^N$ , such that:

$$\mathcal{M} = \bigcup_{i=1}^N c_i, \quad c_i \cap c_j = \emptyset \quad \forall i \neq j \quad (1)$$

For any pair of topologically adjacent cells  $c_i, c_j \in \mathcal{C}$ , their shared Voronoi boundary is denoted by  $\Gamma_{ij} = \partial c_i \cap \partial c_j$ .

| Cell $c_i$ (Origin)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Cell $c_j$ (Neighbor)                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• State: <math>M_w(i), M_c(i), U(i)</math></li> <li>• Potential: <math>\psi_i, T_i, z_i</math></li> <li>• Area: <math>A_i</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• State: <math>M_w(j), M_c(j), U(j)</math></li> <li>• Potential: <math>\psi_j, T_j, z_j</math></li> <li>• Area: <math>A_j</math></li> </ul> |
| <p style="text-align: center;"><b>Shared Boundary Interface</b> <math>M_{ij} = \partial c_i \cap \partial c_j</math></p> <ul style="list-style-type: none"> <li>• Edge length: <math>L_{ij}</math></li> <li>• Centroid distance: <math>d_{ij}</math></li> <li>• Contact areas: <math>A_{ij}^{\text{atm}}, A_{ij}^{\text{sub}}</math></li> <li>• Normal vector: <math>\hat{n}_{ij}</math></li> <li>• Geometric conductance: <math>\gamma_{ij} = L_{ij}/d_{ij}</math></li> <li>• Slope: <math>S_{ij} = (z_j - z_i)/d_{ij}</math></li> </ul> |                                                                                                                                                                                    |

Figure 1: Interface metric topology parameterizing boundary transfer between adjacent cells  $c_i$  and  $c_j$ .

The integral form of a conservation equation for an extensive scalar field  $\psi$  over control volume  $c_i$  is given by:

$$\frac{\partial}{\partial t} \int_{c_i} \psi dV + \sum_{j \in \mathcal{N}(i)} \int_{\Gamma_{ij}} \mathbf{J}_\psi \cdot \hat{n}_{ij} d\Gamma = \int_{c_i} \sigma_\psi dV \quad (2)$$

where  $\mathcal{N}(i)$  is the topological neighbor set of cell  $c_i$ ,  $\hat{n}_{ij}$  is the outward-pointing unit normal on  $\Gamma_{ij}$ , and  $\mathbf{J}_\psi$  is the total flux density vector.

## 3 Metric Tensor Specification: H3CellInterfaceMetrics

To convert boundary integrals into computationally tractable operators, we define the geometric metric tuple  $M_{ij}$  for each directed interface:

$$M_{ij} = \left( L_{ij}, d_{ij}, \theta_{ij}, \hat{n}_{ij}, A_{ij}^{\text{atm}}, A_{ij}^{\text{sub}}, S_{ij}, \gamma_{ij} \right) \quad (3)$$

### 3.1 Metric Invariants

The components of  $M_{ij}$  satisfy strict geometric and topological symmetries:

$$L_{ij} = L_{ji} > 0 \quad (\text{Edge Length Reciprocity}) \quad (4)$$

$$d_{ij} = d_{ji} > 0 \quad (\text{Centroid Distance Reciprocity}) \quad (5)$$

$$\hat{n}_{ij} = -\hat{n}_{ji} \quad (\text{Unit Normal Antisymmetry}) \quad (6)$$

$$S_{ij} = \frac{z_j - z_i}{d_{ij}} = -S_{ji} \quad (\text{Topographic Slope Antisymmetry}) \quad (7)$$

$$\gamma_{ij} = \frac{L_{ij}}{d_{ij}} = \gamma_{ji} > 0 \quad (\text{Dimensionless Conductance Invariance}) \quad (8)$$

### 3.2 Software Implementation Contract

The interface contract is defined in TypeScript within `src/spatial/h3_types.ts`:

```

1 export interface H3CellInterfaceMetrics {
2   readonly originIndex: string;
3   readonly neighborIndex: string;
4   readonly sharedEdgeLengthMeters: number;
5   readonly centroidDistanceMeters: number;
6   readonly bearingRadians: number;
7   readonly normalVector: readonly [number, number, number];
8   readonly atmosphericContactAreaM2: number;
9   readonly subterraneanContactAreaM2: number;
10  readonly topographicSlope: number;
11  readonly geometricConductance: number;
12 }

```

## 4 Coupled Transport and Dissipation Dynamics

### 4.1 Subsurface Hydrodynamic Flux

Saturated and unsaturated porous matrix flow across the subterranean contact area  $A_{ij}^{\text{sub}}$  is parameterized via the Darcy-Richards equation:

$$q_{\text{sub},i \rightarrow j} = -K_{\text{eff}} \left( \frac{\psi_j - \psi_i}{d_{ij}} + S_{ij} \right) \quad (9)$$

where  $K_{\text{eff}} = \sqrt{K_i K_j}$  is the harmonic or geometric mean hydraulic conductivity, and  $\psi$  represents matrix potential. The mass flow rate  $J_{w,\text{sub}}$  [kg · s<sup>-1</sup>] is:

$$J_{w,\text{sub}}^{(i \rightarrow j)} = \rho_w A_{ij}^{\text{sub}} q_{\text{sub},i \rightarrow j} \quad (10)$$

### 4.2 Surface Runoff (Diffusive Wave Overland Flow)

Overland hydraulic flux is driven by the surface energy head  $\mathcal{H}_i = z_i + h_{w,i}$ . The Saint-Venant diffusive wave formulation yields:

$$u_{\text{surf},i \rightarrow j} = -\text{sgn}(\nabla \mathcal{H}_{ij}) \frac{1}{n_{\text{manning}}} \bar{h}_w^{2/3} \left| \frac{\mathcal{H}_j - \mathcal{H}_i}{d_{ij}} \right|^{1/2} \quad (11)$$

operating across the surface cross-section  $A_{ij}^{\text{surf}} = L_{ij} \bar{h}_w$ , where  $\bar{h}_w = \frac{1}{2}(h_{w,i} + h_{w,j})$ . The surface mass flux is:

$$J_{w,\text{surf}}^{(i \rightarrow j)} = \rho_w L_{ij} \bar{h}_w u_{\text{surf},i \rightarrow j} \quad (12)$$

### 4.3 Atmospheric Boundary-Layer Enthalpy Transfer

Thermal transport through the planetary boundary layer cross-section  $A_{ij}^{\text{atm}}$  decomposes into advective and turbulent diffusive components:

$$J_{H,\text{adv}}^{(i \rightarrow j)} = \rho_{\text{air}} c_p A_{ij}^{\text{atm}} \left( \frac{\vec{u}_i + \vec{u}_j}{2} \cdot \hat{n}_{ij} \right) T_{ij}^* \quad (13)$$

$$J_{H,\text{diff}}^{(i \rightarrow j)} = -\rho_{\text{air}} c_p K_T \left( \frac{A_{ij}^{\text{atm}}}{d_{ij}} \right) (T_j - T_i) \quad (14)$$

where  $T_{ij}^*$  is evaluated using first-order upwind discretization according to the sign of the normal wind velocity component  $u_{n,ij}$ .

### 4.4 Upwind Advective Solute & Carbon Transport

For dissolved nutrients ( $N, P$ ), carbon fractions (DOC, DIC), and dissolved oxygen ( $DO$ ), mass fluxes are coupled to the total water flux  $J_w^{(i \rightarrow j)} = J_{w,\text{sub}}^{(i \rightarrow j)} + J_{w,\text{surf}}^{(i \rightarrow j)}$ :

$$J_k^{(i \rightarrow j)} = C_{k,ij}^* J_w^{(i \rightarrow j)}, \quad C_{k,ij}^* = \begin{cases} C_{k,i}, & \text{if } J_w^{(i \rightarrow j)} \geq 0 \\ C_{k,j}, & \text{if } J_w^{(i \rightarrow j)} < 0 \end{cases} \quad (15)$$

## 5 Thermodynamic Invariance Proofs

**Theorem 1** (First Law Conservation Invariance). *Let  $\Phi_{i \rightarrow j}(S)$  denote the total boundary transfer of any extensive stock  $S \in \{M_w, M_c, U\}$  across  $M_{ij}$  over time interval  $\Delta t$ . Under the metric symmetries of  $M_{ij}$ , the exchange is strictly skew-symmetric:*

$$\Phi_{i \rightarrow j}(S) = -\Phi_{j \rightarrow i}(S) \quad (16)$$

and globally conservative over the manifold  $\mathcal{M}$ :

$$\sum_{c_i \in \mathcal{C}} \sum_{j \in \mathcal{N}(i)} \Phi_{i \rightarrow j}(S) = 0 \quad (17)$$

*Proof.* Consider the subsurface water flux  $J_{w,\text{sub}}^{(i \rightarrow j)}$ . By definition:

$$J_{w,\text{sub}}^{(j \rightarrow i)} = \rho_w A_{ji}^{\text{sub}} \left( -K_{\text{eff}} \left( \frac{\psi_i - \psi_j}{d_{ji}} + S_{ji} \right) \right) \quad (18)$$

Substituting the metric invariants  $A_{ji}^{\text{sub}} = A_{ij}^{\text{sub}}$ ,  $d_{ji} = d_{ij}$ , and  $S_{ji} = -S_{ij}$ :

$$J_{w,\text{sub}}^{(j \rightarrow i)} = \rho_w A_{ij}^{\text{sub}} \left( -K_{\text{eff}} \left( -\frac{\psi_j - \psi_i}{d_{ij}} - S_{ij} \right) \right) \quad (19)$$

$$= -\rho_w A_{ij}^{\text{sub}} \left( -K_{\text{eff}} \left( \frac{\psi_j - \psi_i}{d_{ij}} + S_{ij} \right) \right) = -J_{w,\text{sub}}^{(i \rightarrow j)} \quad (20)$$

Identical antisymmetry holds for surface runoff  $J_{w,\text{surf}}$ , advective solute transfer  $J_k$ , and diffusive enthalpy  $J_{H,\text{diff}}$ . For the global manifold:

$$\sum_{c_i \in \mathcal{C}} \sum_{j \in \mathcal{N}(i)} \Phi_{i \rightarrow j}(S) = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} (\Phi_{i \rightarrow j}(S) + \Phi_{j \rightarrow i}(S)) = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} (0) = 0 \quad (21)$$

where  $\mathcal{E}$  is the set of all undirected unique cell interfaces in  $\mathcal{M}$ .  $\square$

**Theorem 2** (Second Law Non-Negative Entropy Production). *The entropy production rate  $\dot{S}_{gen,ij}$  associated with thermal diffusion across interface  $M_{ij}$  is non-negative:*

$$\dot{S}_{gen,ij} \geq 0 \quad (22)$$

*Proof.* The local entropy generation rate for heat transfer between two thermal reservoirs at temperatures  $T_i$  and  $T_j$  across interface  $M_{ij}$  is given by:

$$\dot{S}_{gen,ij} = J_{H,diff}^{(i \rightarrow j)} \left( \frac{1}{T_j} - \frac{1}{T_i} \right) \quad (23)$$

Substituting the Fourier-Laplacian heat flux:

$$\dot{S}_{gen,ij} = \left( -\rho_{air} c_p K_T \frac{A_{ij}^{atm}}{d_{ij}} (T_j - T_i) \right) \left( \frac{T_i - T_j}{T_i T_j} \right) \quad (24)$$

$$= \rho_{air} c_p K_T \left( \frac{A_{ij}^{atm}}{d_{ij}} \right) \frac{(T_j - T_i)^2}{T_i T_j} \quad (25)$$

Since  $\rho_{air} > 0$ ,  $c_p > 0$ ,  $K_T > 0$ ,  $A_{ij}^{atm} > 0$ ,  $d_{ij} > 0$ , and absolute temperatures  $T_i, T_j > 0$ , every term is strictly positive. Therefore:

$$\dot{S}_{gen,ij} \geq 0 \quad (26)$$

with  $\dot{S}_{gen,ij} = 0$  if and only if  $T_i = T_j$ .  $\square$

## 6 Software Architecture and Monad Pipeline

The introduction of `H3CellInterfaceMetrics` structures the transport execution graph of the simulation engine:

Within `SpatialMonad`, the discrete 2D Laplace-Beltrami operator  $\nabla^2 \phi$  for any conservative scalar  $\phi$  is discretized via the geometric conductance  $\gamma_{ij}$ :

$$\nabla^2 \phi_i \approx \frac{1}{A_i} \sum_{j \in \mathcal{N}(i)} \gamma_{ij} (\phi_j - \phi_i) = \frac{1}{A_i} \sum_{j \in \mathcal{N}(i)} \frac{L_{ij}}{d_{ij}} (\phi_j - \phi_i) \quad (27)$$

This formulation preserves Voronoi flux orthogonality and eliminates spatial grid anisotropy across aperture-7 hexagonal transitions.

## 7 Verification and Numerical Integrity

Unit and property-based test suites in `tests/sprint_051.test.ts` validate the implementation against the formal invariant specifications:

| Invariant ID | Physical Target             | Mathematical Form                                     | Verification Tolerance           |
|--------------|-----------------------------|-------------------------------------------------------|----------------------------------|
| INV-051-A    | Voronoi Edge Symmetry       | $L_{ij} = L_{ji}$                                     | Precision $\epsilon < 10^{-12}$  |
| INV-051-B    | Centroid Distance Symmetry  | $d_{ij} = d_{ji}$                                     | Precision $\epsilon < 10^{-12}$  |
| INV-051-C    | Facet Normal Inversion      | $\hat{n}_{ij} = -\hat{n}_{ji}$                        | Norm error $< 10^{-9}$           |
| INV-051-D    | Topographic Slope Inversion | $S_{ij} = -S_{ji}$                                    | Residual $< 10^{-9}$             |
| INV-051-E    | Conductance Positivity      | $\gamma_{ij} = L_{ij}/d_{ij} > 0$                     | Strict inequality                |
| INV-051-F    | Conservative Flux Pair      | $\Phi_{i \rightarrow j} + \Phi_{j \rightarrow i} = 0$ | Machine zero ( $\leq 10^{-14}$ ) |
| INV-051-G    | Second Law Positivity       | $\dot{S}_{gen} \geq 0$                                | Absolute $\geq -10^{-12}$        |

Table 1: Automated property invariants enforcing thermodynamic and geometric consistency.

